

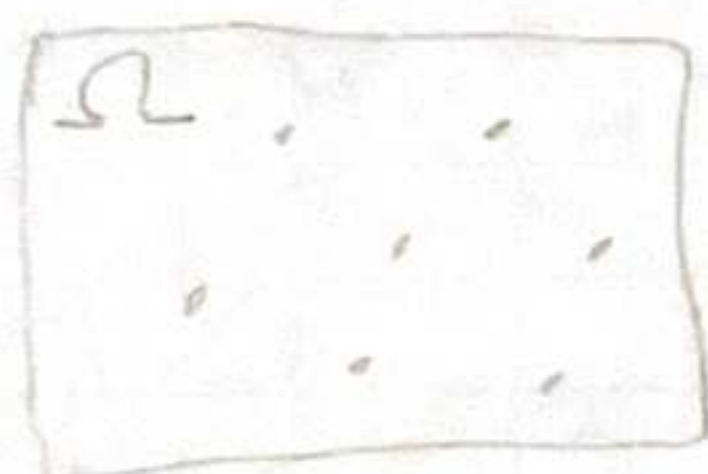
L1.1 • Lecture 1: Probability models & axioms

L1.2 • Sample Space

→ Putting together a probabilistic model requires

- Two Steps
 1. Describe possible outcomes
 2. Describe beliefs about likelihood of outcomes

① List (set) of possible outcomes, Ω



- List must be:
 - Mutually exclusive
 - Collectively exhaustive
 - At the "right" granularity.

→ 1 outcome

→ outcome within sample set.

Example: Flip a coin Ω



Let's say look outside at the weather

- H & rains Ω
- H & no rain
- T & rains
- T & no rain

↳ Valid from outcome perspective but to purely flip coin, rain may not be relevant.

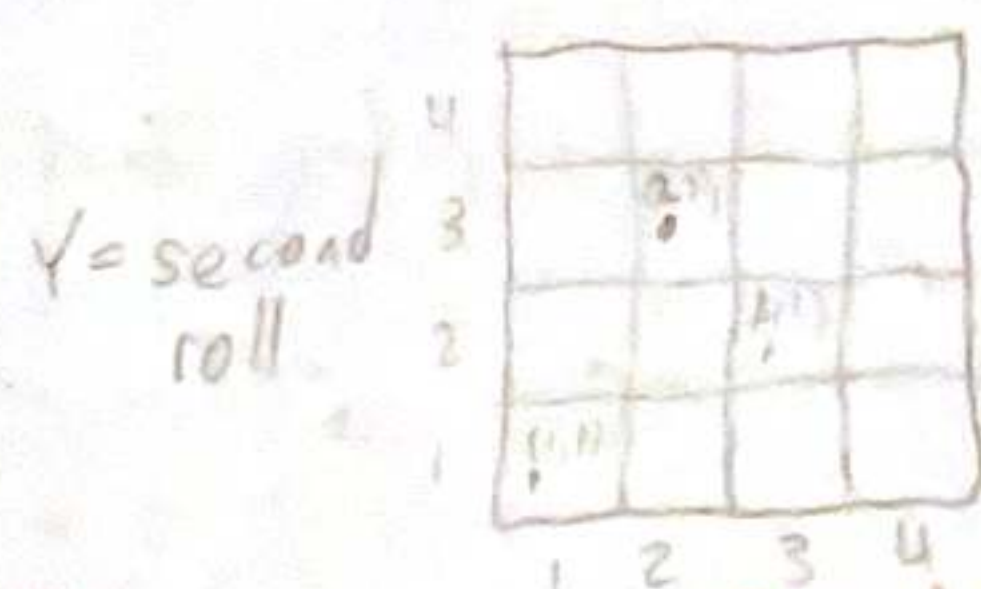
↳ Unless want to capture relevance of weather.

L1.3 • Sample Space Examples

↳ Could be discrete, continuous, finite, infinite.

→ Example (discrete / finite)

- Two rolls of a tetrahedral die

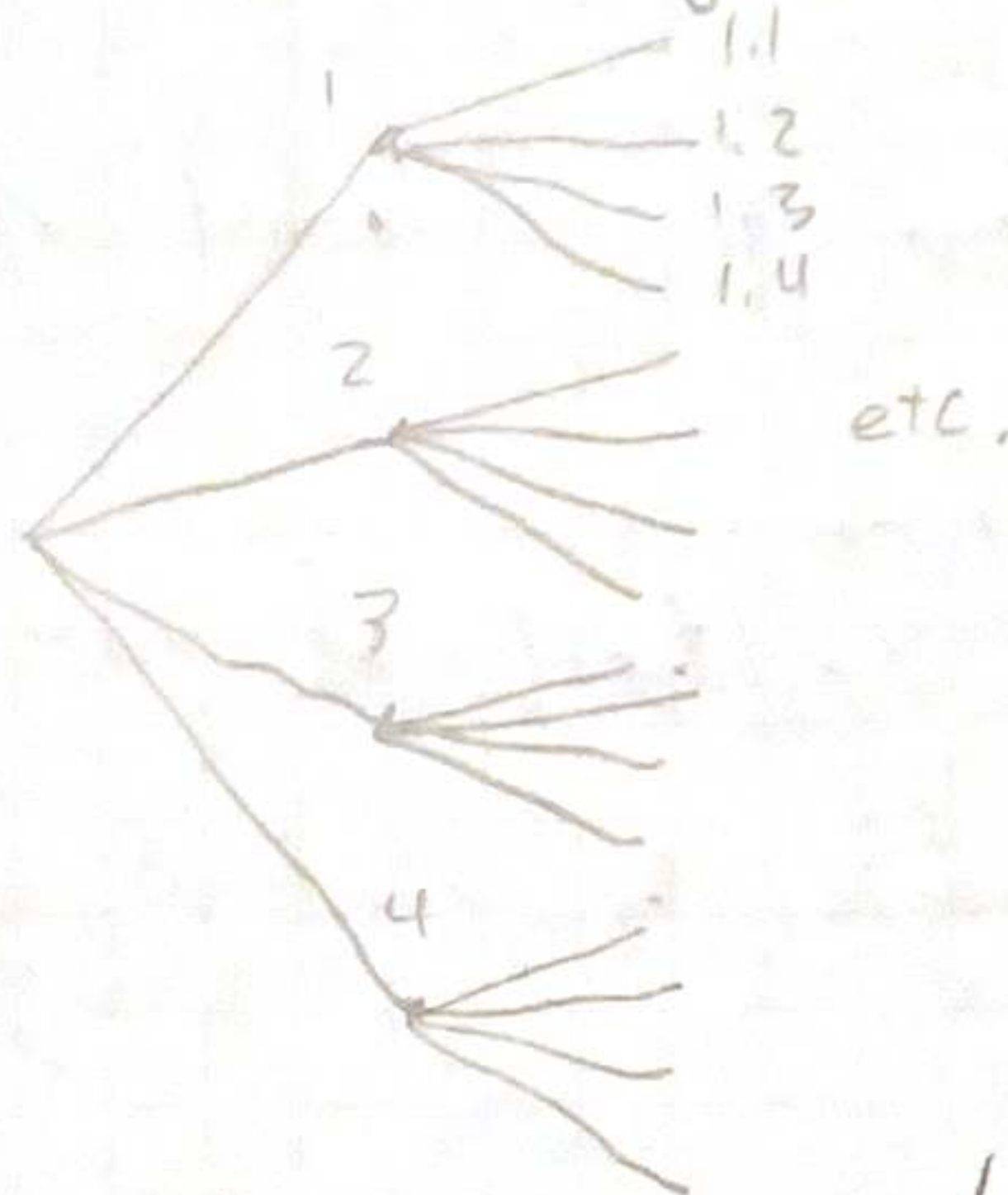


Note: (2, 3) & (3, 2) are distinct outcomes.

laborn jitziladung a rot bringer 110 ← amoxA pln0 a

• Sequential description

Two dice rolls again



Number of leaves = 16
 ↳ Same as 4×4 grid approach.

Tree → used for sequential description

• Sample Space - Continuous Example.

(x, y) such that $0 \leq x, y \leq 1$

L1.4 • Probability Axioms

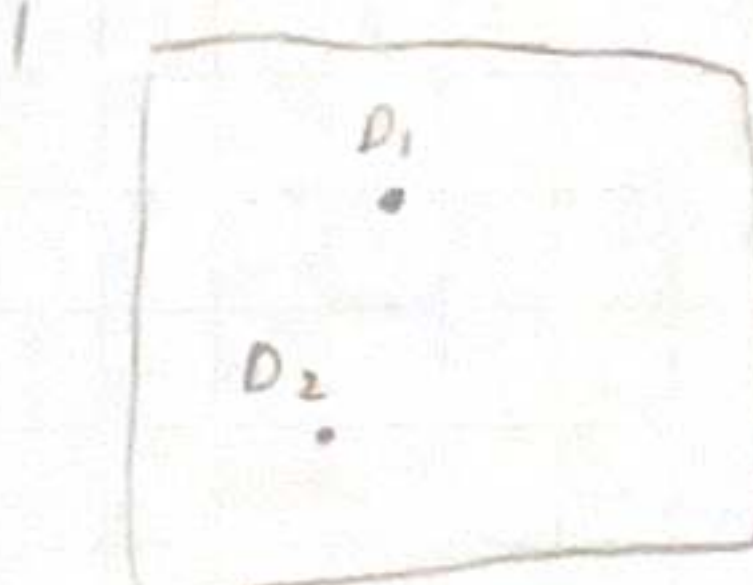
- Event: a subset of the sample space.
 — Probability assigned to events.

Continuous Example: Throwing darts

• Axioms:

⊖ Nonnegativity: $P(A) \geq 0$

⊕ Normalization: $P(\Omega) = 1$
 ↳ Definition of sample space includes all possibilities.



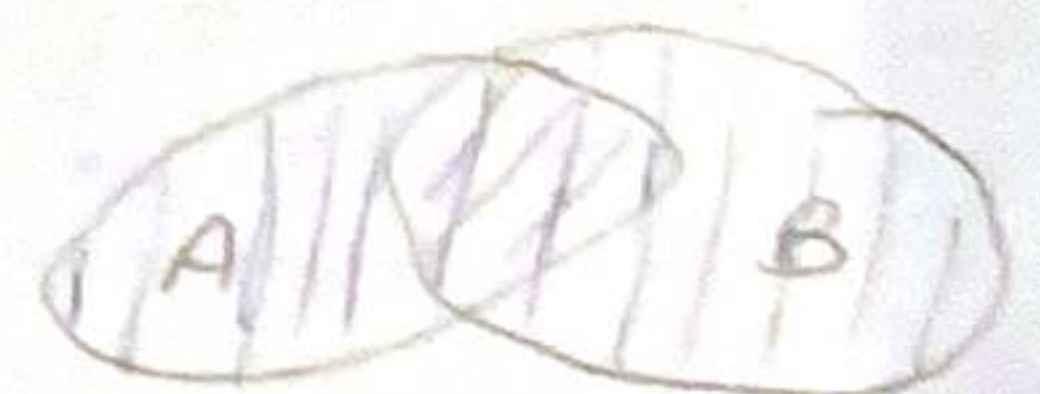
→ Prob(dart hits exact centre) = 0
 ↳ Infinite precision impossible.



$P(A) = \underline{\hspace{2cm}}$
 ↳ Event A → subset
 ↳ Possible to achieve.

⊕ (Finite) additivity: (to be strengthened later)

If $A \cap B = \emptyset$, then $P(A \cup B) = P(A) + P(B)$



↳ $\emptyset$ used to represent empty set.

↳ Can add prob of two sets when those sets are disjoint.

→ Only Axioms → all required for a probabilistic model.

Properties of Probability Axioms

Axioms

$$(a) \quad P(A) \geq 0$$

$$(b) \quad P(\Omega) = 1$$

Consequences

$$P(A) \leq 1$$

$$P(\emptyset) = 0$$

↳ empty set

For disjoint events:

$$(c) \quad P(A \cup B) = P(A) + P(B)$$

$$P(A) + P(A^c) = 1$$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

and similarly for k disjoint events

$$P(\{s_1, s_2, \dots, s_k\}) = P(\{s_1\}) + \dots + P(\{s_k\})$$

$$= P(s_1) + \dots + P(s_k)$$

$$A \cup A^c = \Omega$$

$$A \cap A^c = \emptyset$$



c - complement.

$\left\{ \begin{array}{l} \cup - \text{union} \\ \cap - \text{intersection} \end{array} \right\}$

$$1 \stackrel{(b)}{=} P(\Omega) = P(A \cup A^c)$$

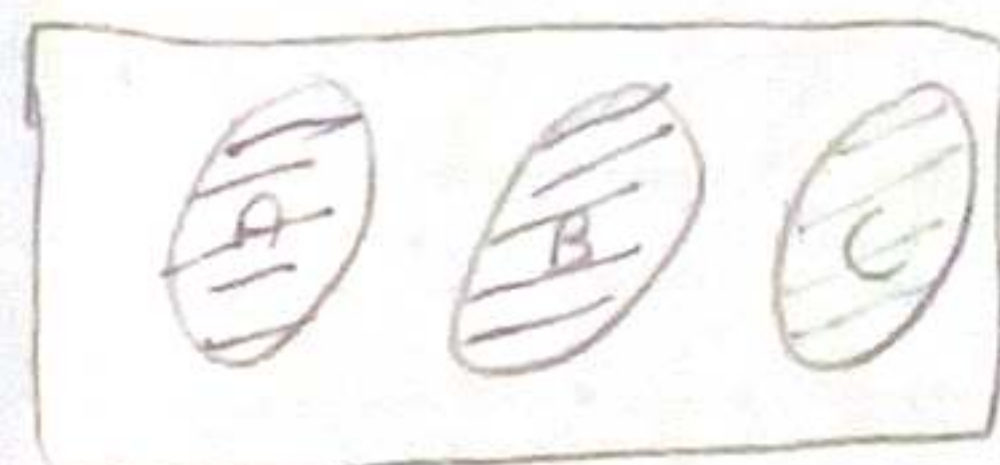
$$\stackrel{(c)}{=} P(A) + P(A^c) \stackrel{(a)}{\leq} 1$$

$$P(A) = 1 - P(A^c) \leq 1$$

$$1 = P(\Omega) + P(\Omega^c)$$

$$1 = (1) + \cancel{P(\emptyset)} \Rightarrow P(\emptyset) = 0$$

A, B, C disjoint: $P(A \cup B \cup C) = P(A) + P(B) + P(C)$
↳ no events in common



$$P(A \cup B \cup C) = P((A \cup B) \cup C) = P(A \cup B) + P(C)$$

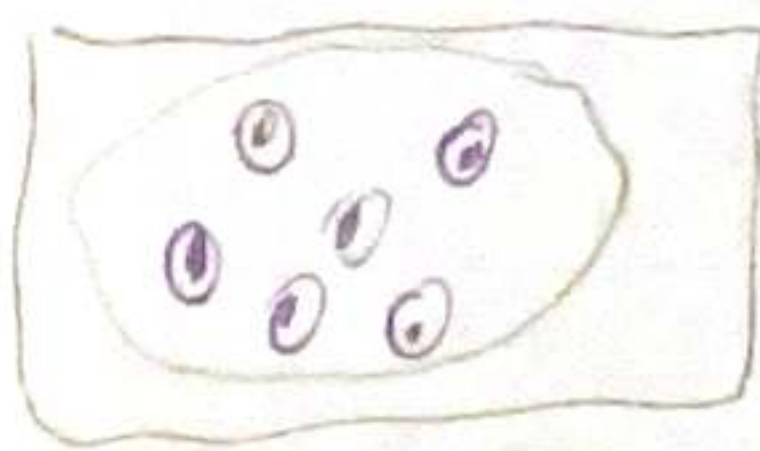
$$= P(A) + P(B) + P(C)$$

If $A_1, \dots, A_k$ disjoint $\Rightarrow P(A_1 \cup \dots \cup A_k) = \sum_{i=1}^k P(A_i)$

$$P(\{s_1, s_2, \dots, s_k\}) = P(\{s_1\} \cup \{s_2\} \cup \dots \cup \{s_k\})$$

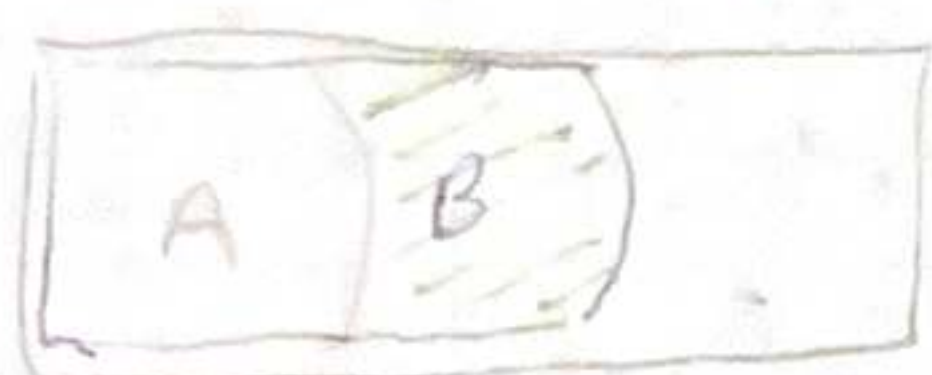
$$= P(\{s_1\}) + \dots + P(\{s_k\})$$

$$= P(s_1) + \dots + P(s_k)$$



L1.6 • More Properties of Probabilities

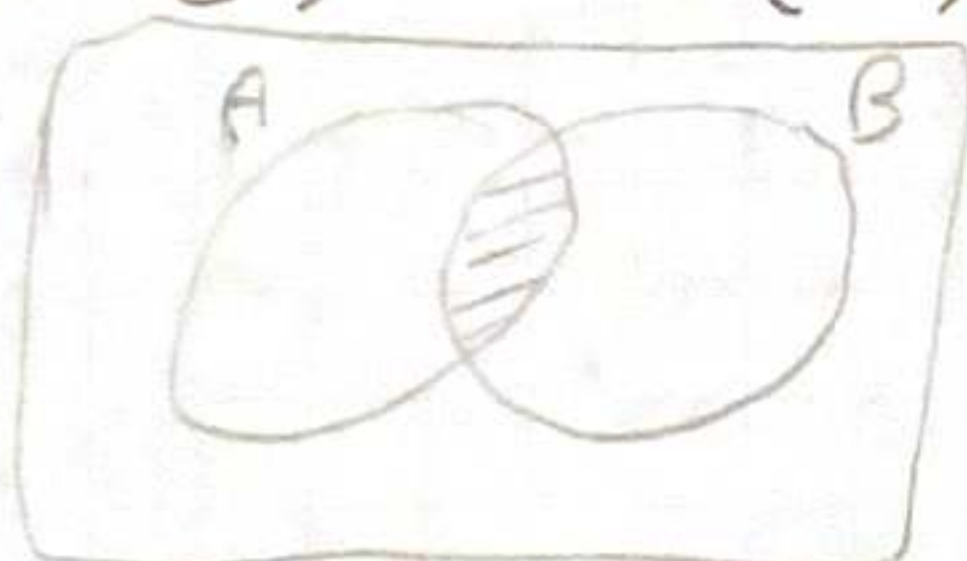
- If $A \subset B$, then $P(A) \leq P(B)$



$$B = A \cup (B \cap A^c)$$

$$P(B) = P(A) + P(B \cap A^c) \geq P(A)$$

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$



$$a = P(A \cap B^c)$$

$$b = P(A \cap B)$$

$$c = P(B \cap A^c)$$

$$\therefore P(A \cup B) = a + b + c$$

$$P(A) + P(B) - P(A \cap B) = (a+b) + (b+c) - b \\ = a + b + c$$

- $P(A \cup B) \leq P(A) + P(B)$ called "Union Bound"

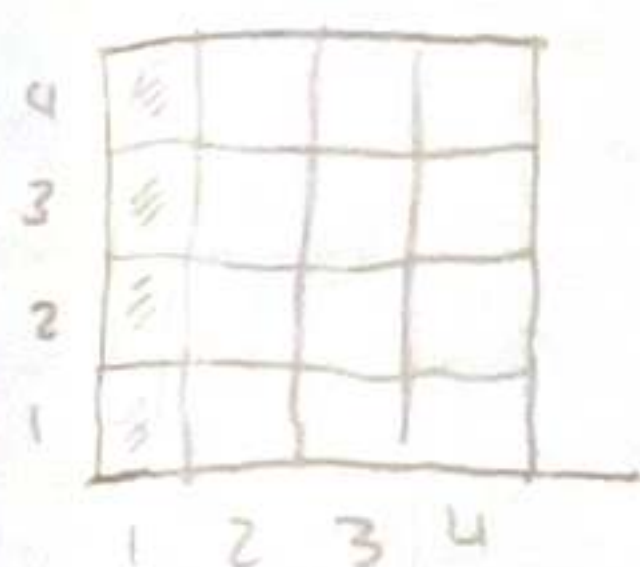
$$P(A \cup B \cup C) = P(A) + P(A^c \cap B) + P(A^c \cap B^c \cap C)$$

$$A \cup B \cup C =$$

$$= A \cup (B \cap A^c) \cup (C \cap A^c \cap B^c)$$

L1.7 • A Discrete Example

- Two rolls of a tetrahedral die
- Let every possible outcome have a probability $1/16$.



$$P(X=1) = 4 \cdot \frac{1}{16} = \frac{1}{4} = 0.25$$

$$\text{Let } Z = \min(X, Y)$$

$$x=2, y=3, Z=2.$$

$$P(Z=4) = 1 \cdot \frac{1}{16} = \frac{1}{16} \quad \text{--- } \{x=4; y=4\}$$

$$P(Z=2) = 5 \cdot \frac{1}{16} = \frac{5}{16} \quad \text{OR } \begin{cases} x=2; y \geq 2 \\ y=2; x \geq 2 \end{cases}$$

Discrete uniform law

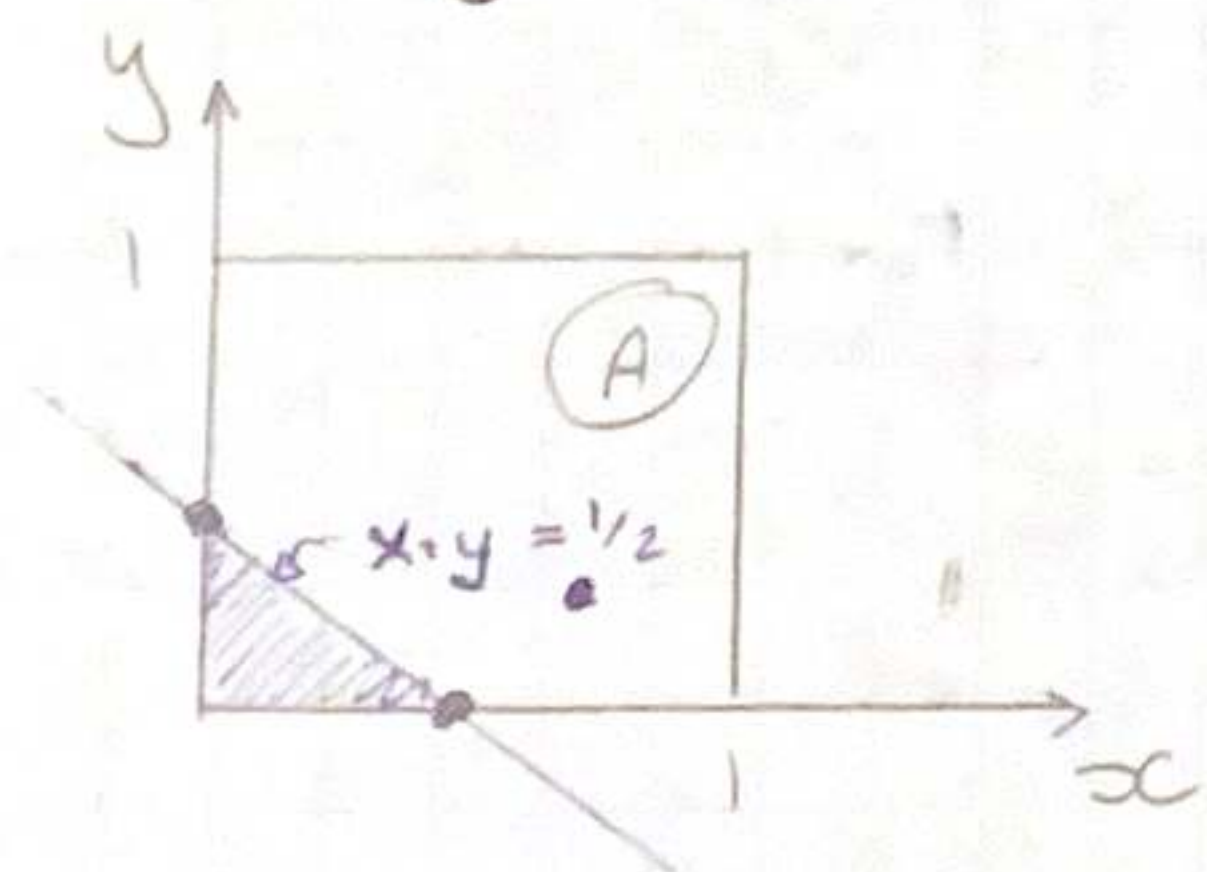
- Assume Ω consists of n equally likely elements.
- Assume A consists of K elements

$$P(A) = K \cdot \frac{1}{n}$$

→ Basically count events as each event is uniform and its discrete.

L1.8 • A Continuous Example throw darts

- (x, y) such that $0 \leq x, y \leq 1$



Uniform probability law: Probability = Area
→ Arbitrary → just for example.

$$P(\{(x, y) \mid x+y \leq \frac{1}{2}\}) = \frac{1}{2} \cdot (\frac{1}{2}) \cdot (\frac{1}{2}) = \frac{1}{8}$$

↳ Area Δ

$$P(\{0.5; 0.3\}) = 0 \quad (\text{Single point's area} \rightarrow \text{points so no area})$$

• Probability calculation steps

- ① Specify the sample space
 - ② Specify a probability law $\rightarrow$ Something relevant to problem of interest.
 - ③ Identify an event of interest.
 - ④ Calculate ...
- $\leadsto$ Systematic approach to building a probability model.

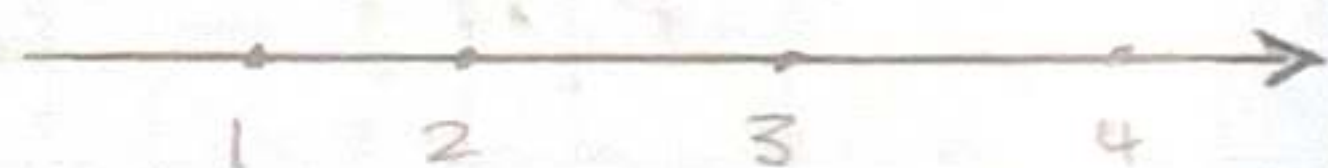
L1.9 • Countable Additivity

Probability calculation $\rightarrow$ discrete but infinite sample space.

① Sample space: $\{1, 2, \dots\}$

② Law:

— We are given $P(n) = \frac{1}{2^n}$, $n = 1, 2, \dots$



Check P sums to 1

$$\sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1}{2} \sum_{n=0}^{\infty} \frac{1}{2^n} = \frac{1}{2} \cdot \frac{1}{1 - (1/2)} = 1 \rightarrow \text{Legit Prob Law}$$

$$\hookrightarrow P(n) = \frac{1}{2^n} \quad \checkmark$$

• $P(\text{outcome is even}) = P(\{2, 4, 6, \dots\})$

$$= P(\{2\} \cup \{4\} \cup \{6\} \cup \dots) = P(2) + P(4) + P(6) + \dots$$

$$= \frac{1}{2^2} + \frac{1}{2^4} + \frac{1}{2^6} + \dots$$

$$= \frac{1}{4} (1 + \frac{1}{4} + \frac{1}{4^2} + \dots) = \frac{1}{4} \cdot \frac{1}{1 - (1/4)} = \frac{1}{3}$$

Additive axiom was for finitely many P s.
Technically not allowed for infinite events.

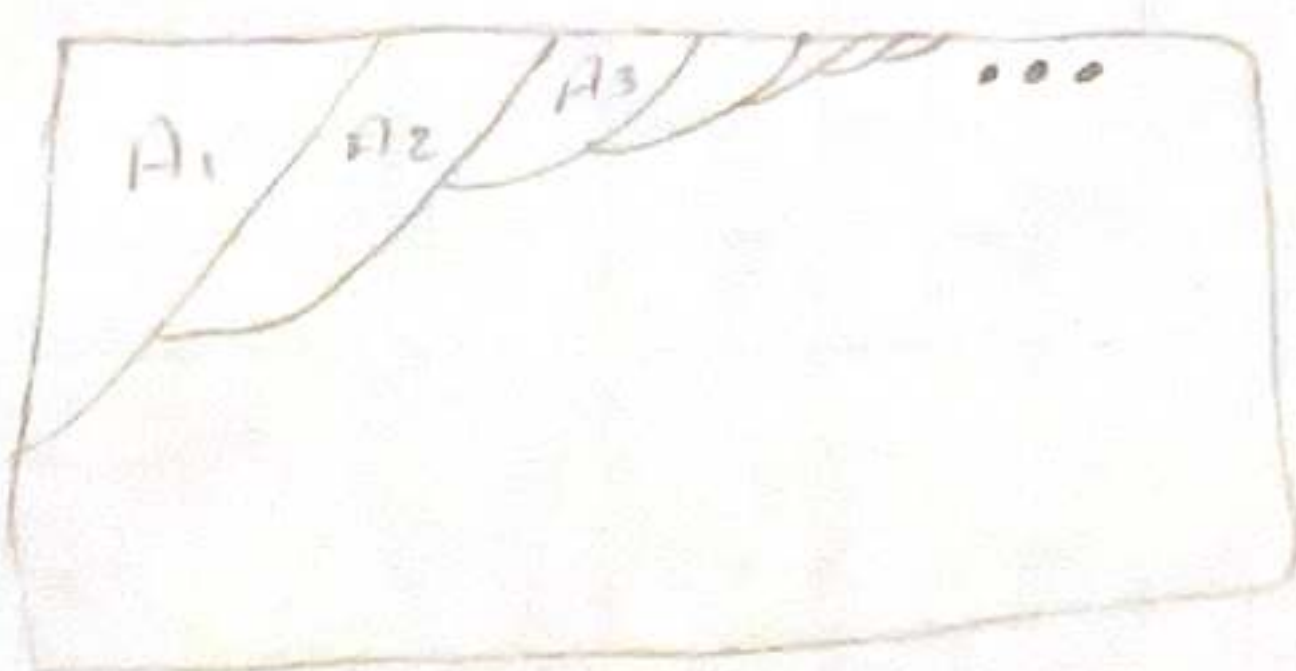
Introduce new Axiom

④ Countable Additivity Axiom

If $A_1, A_2, A_3, \dots$ is an infinite sequence of disjoint events,

then $P(A_1 \cup A_2 \cup A_3 \cup \dots) = P(A_1) + P(A_2) + P(A_3) + \dots$

e.g.

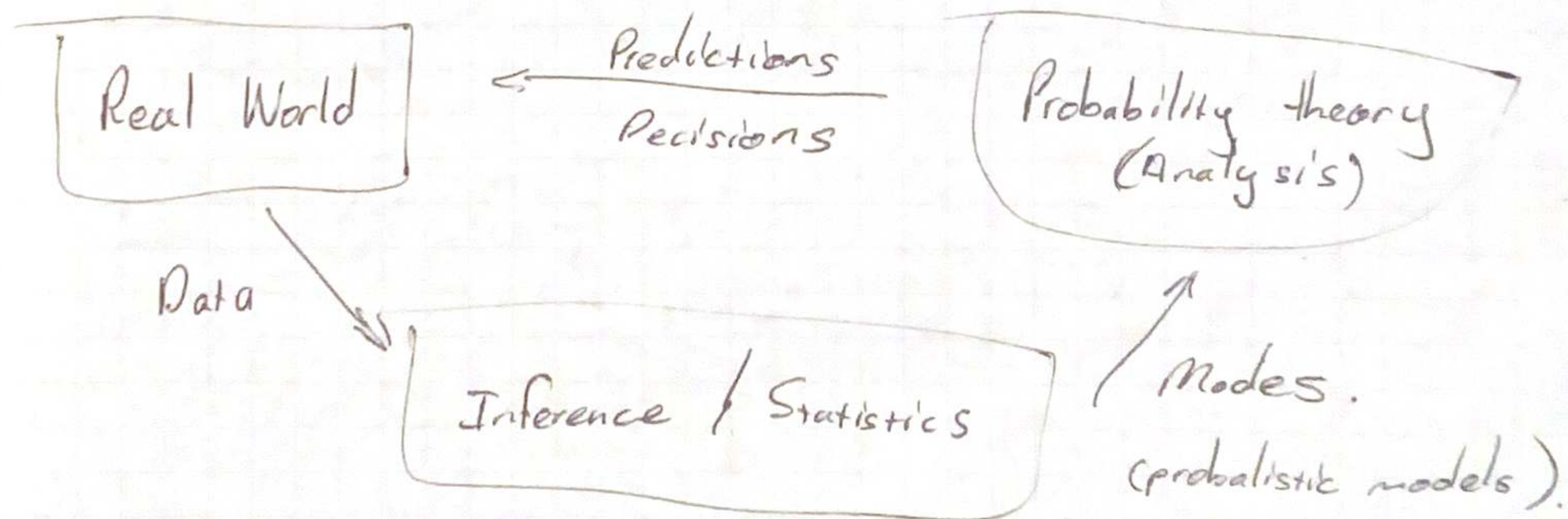


• Additivity holds only for "countable" sequences of events

• The unit square (similarly, the real line, etc.) is not countable (its elements cannot be arranged in a sequence).

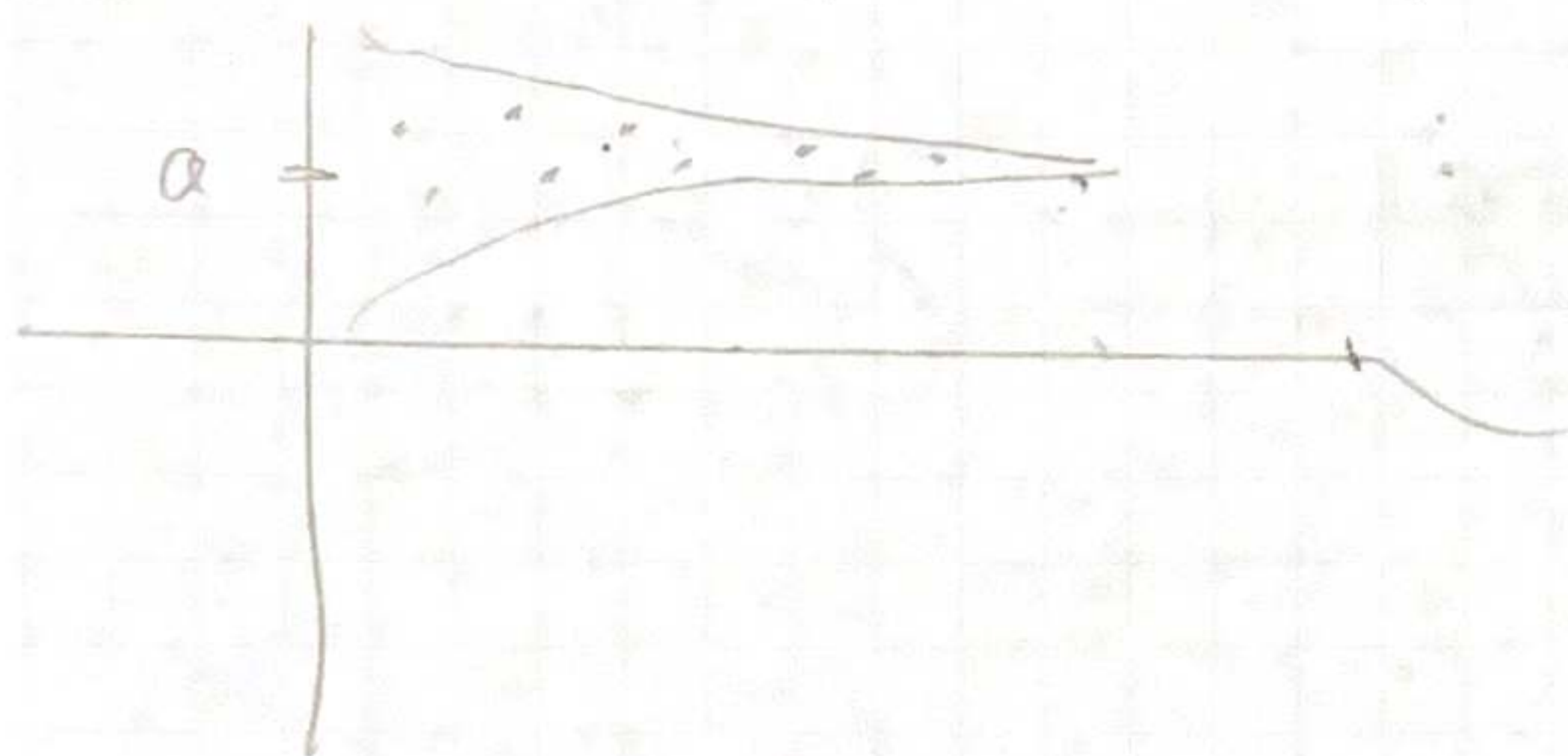
L1.10 • Interpretations & Uses of Probability

- A narrow view : a branch of maths
 - Axioms $\Rightarrow$ theorems "Thn: " "Frequency" of event A "is" $P(A)$
- Are probabilities frequencies?
 - $P(\text{coins toss yields heads}) = 1/2$
 - $P(\text{the president of ... will be ~~reached~~ reelected}) = 0.7$
- Probabilities are often interpreted as:
 - description of beliefs
 - betting preferences.
- The role of probability theory.
 - A framework for analyzing phenomena with uncertain outcomes.
 - Rules for consistent reasoning
 - Used for predictions and decisions.



• When does a sequence converge?

- If $a_i \leq a_{i+1}$, for all i , then either:
 - the sequence "converges to ∞ "
 - the sequence converges to some real number a .
- If $|a_i - a| \leq b_i$, for all i , and $b_i \rightarrow 0$, then $a_i \rightarrow a$.



• Infinite Series

$$\sum_{i=1}^{\infty} a_i = \lim_{n \rightarrow \infty} \sum_{i=1}^n a_i, \text{ provided the limit exists.}$$

- If $a_i \geq 0$: limit exists.
- if terms a_i do not all have the same sign:
 - limit need not exist
 - limit may exist but be different if we sum in a different order

- Fact: limit exists and independent of order of summation
if $\sum_{i=1}^{\infty} |a_i| < \infty$

• Geometric Series

$$S = \sum_{i=0}^{\infty} \alpha^i = 1 + \alpha + \alpha^2 + \dots \quad |\alpha| < 1$$

$$(1 - \alpha)(1 + \alpha + \dots + \alpha^n) = 1 - \alpha^{n+1}$$

$n \rightarrow \infty$

$$(1 - \alpha)S = 1$$

$$S = \frac{1}{1 - \alpha}$$

$\rightarrow$ goes to zero $\rightarrow \infty$.

$$\sum_{i=0}^{\infty} \alpha^i = 1 + \alpha + \alpha^2 + \dots = \frac{1}{1 - \alpha} \quad |\alpha| < 1$$

• Maths Review

• Set Properties

$$S \cup T = T \cup S$$

$$S \cap (T \cup U) = (S \cap T) \cup (S \cap U)$$

$$(S^c)^c = S$$

$$S \cup \Omega = \Omega$$

$$S \cup (T \cap U) = (S \cup T) \cap (S \cup U)$$

$$S \cap (T \cup U) = (S \cap T) \cup (S \cap U)$$

$$S \cap S^c = \emptyset$$

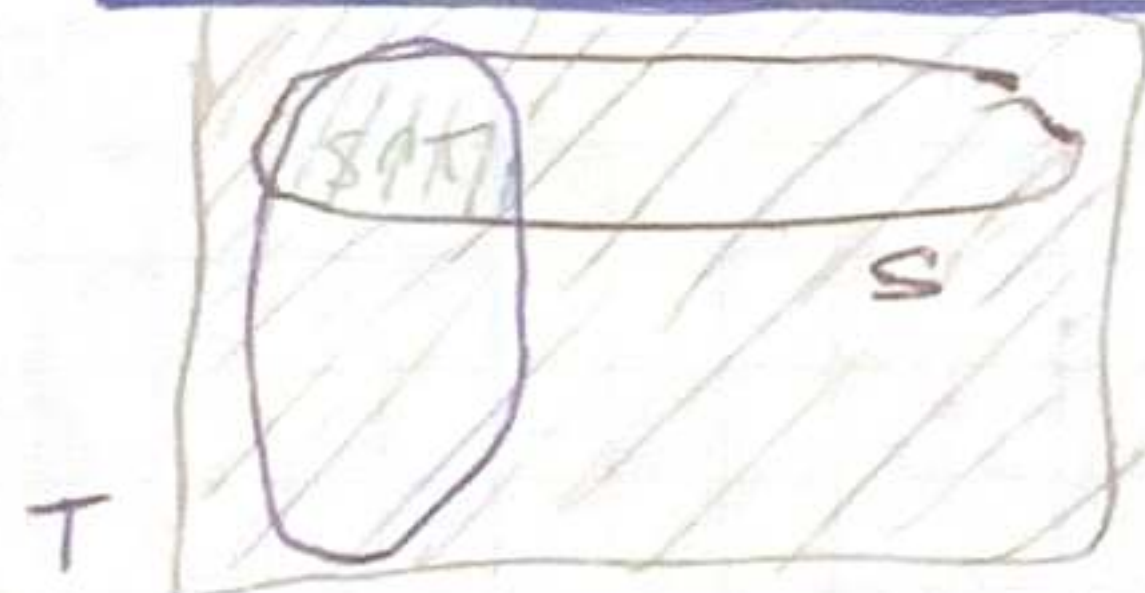
$$S \cap \Omega = S$$

• De Morgan's laws

(Often used to move between
Unions & Intersections)

$$\textcircled{1} \quad \underline{(S \cap T)^c = S^c \cup T^c}$$

$$(\bigcap S_n)^c = \bigcup S_n^c$$



$$\textcircled{2} \quad (S^c \cap T^c)^c = S \cup T$$

$$\underline{S^c \cap T^c = (S \cup T)^c} \quad (\bigcup S_n)^c = \bigcap S_n^c$$

• Sequences & their Limits

$a_1, a_2, a_3, \dots$
sequence $a_i, \{a_i\}$

$i \in \mathbb{N} = \{1, 2, 3, \dots\}$

function $f: \mathbb{N} \rightarrow S$
 $f(i) = a_i$

$$a_i \xrightarrow{i \rightarrow \infty} a$$

$$\lim_{i \rightarrow \infty} a_i = a$$

For any $\epsilon > 0$, there exists i_0 such that if $i \geq i_0$, then $|a_i - a| < \epsilon$

($\epsilon \rightarrow$ epsilon)

• Introduction to Probability · Lecture 2

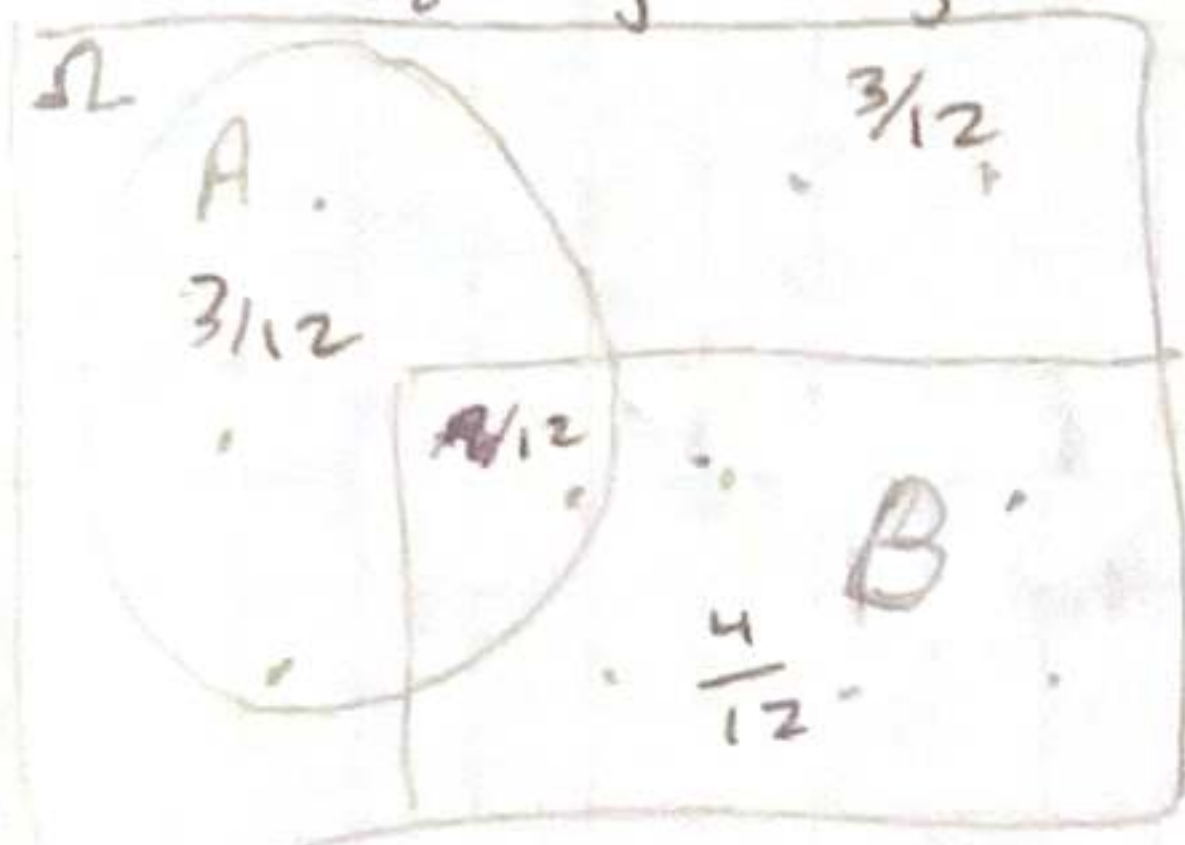
L2.1 • Lecture 2 : Conditioning and Bayes Rule

• Three tools

1. Multiplication Rule
2. Total probability theorem.
3. Bayes' rule ($\rightarrow$ inference)

L2.2 • Conditioning

$\rightarrow$ 12 equally likely outcomes.



$$P(A) = \frac{3}{12} \quad P(B) = \frac{6}{12}$$

If told B occurred:

$P(A|B) \Rightarrow$ Probability A given B.

$$P(A|B) = \frac{2}{6} = \frac{1}{3} \quad P(B|B) = 1$$

• Direction of conditional probability

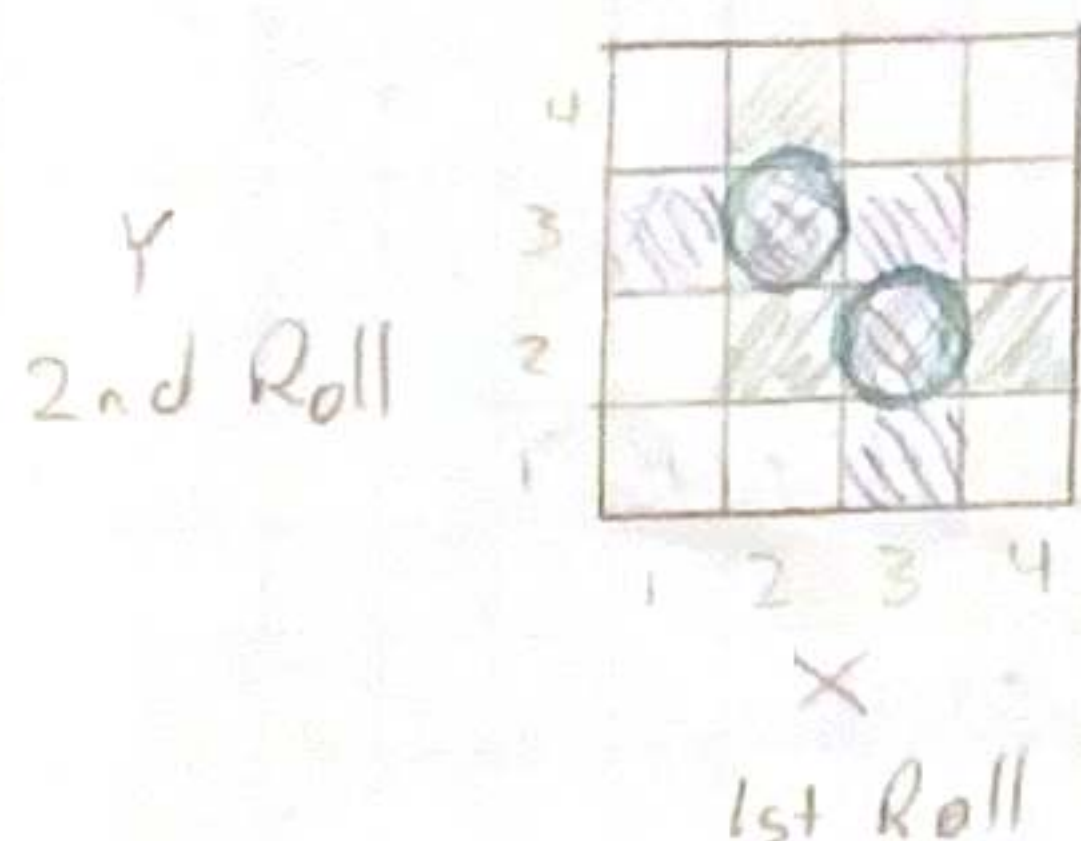
$P(A|B)$ = "probability of A, given that B occurred"

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{defined only when } P(B) > 0.$$

e.g. $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{2/12}{6/12} = \frac{1}{3}$ ✓

~~defined~~

L2.3 • Die Roll Example



Let B be event: $\min(X, Y) = 2$

Let $M = \max(X, Y)$

$$P(M = 1 | B) = 0$$

$$P(M = 3 | B) = \frac{P(M = 3 \text{ and } B)}{P(B)}$$

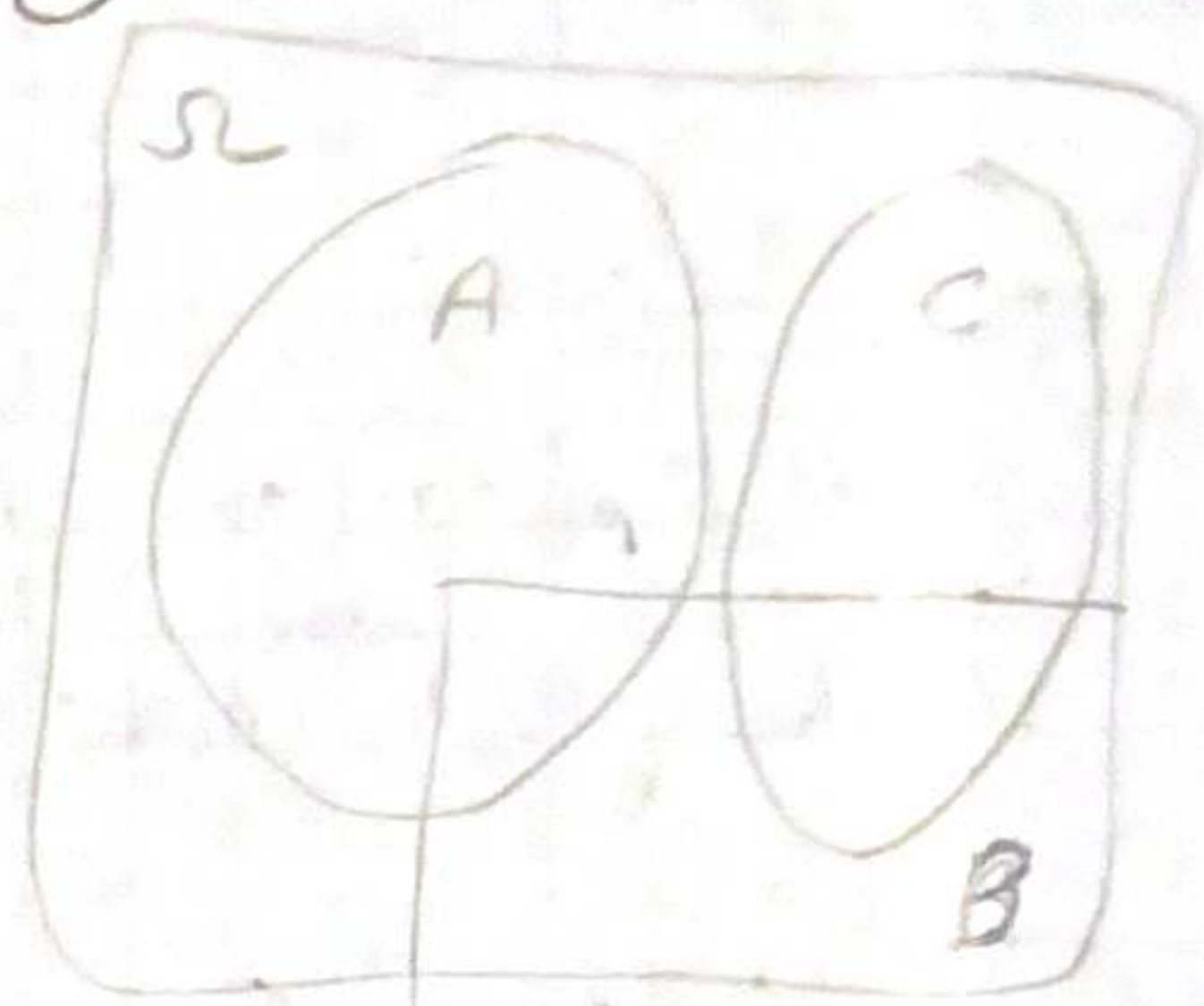
$$= \frac{(2/16)}{(5/16)} = \frac{2}{5}$$

8 events possible give B occurred.

2.4 • Conditional Probabilities Obey the Same Axioms

• $P(A|B) \geq 0$ assuming $P(B) > 0$

• $P(\Omega|B) = \frac{P(\Omega \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1$



If $A \cap C = \emptyset$, then:

• $P(A \cup C|B) = P(A|B) + P(C|B)$

= $\frac{P((A \cup C) \cap B)}{P(B)} = \frac{P((A \cap B) \cup (C \cap B))}{P(B)}$

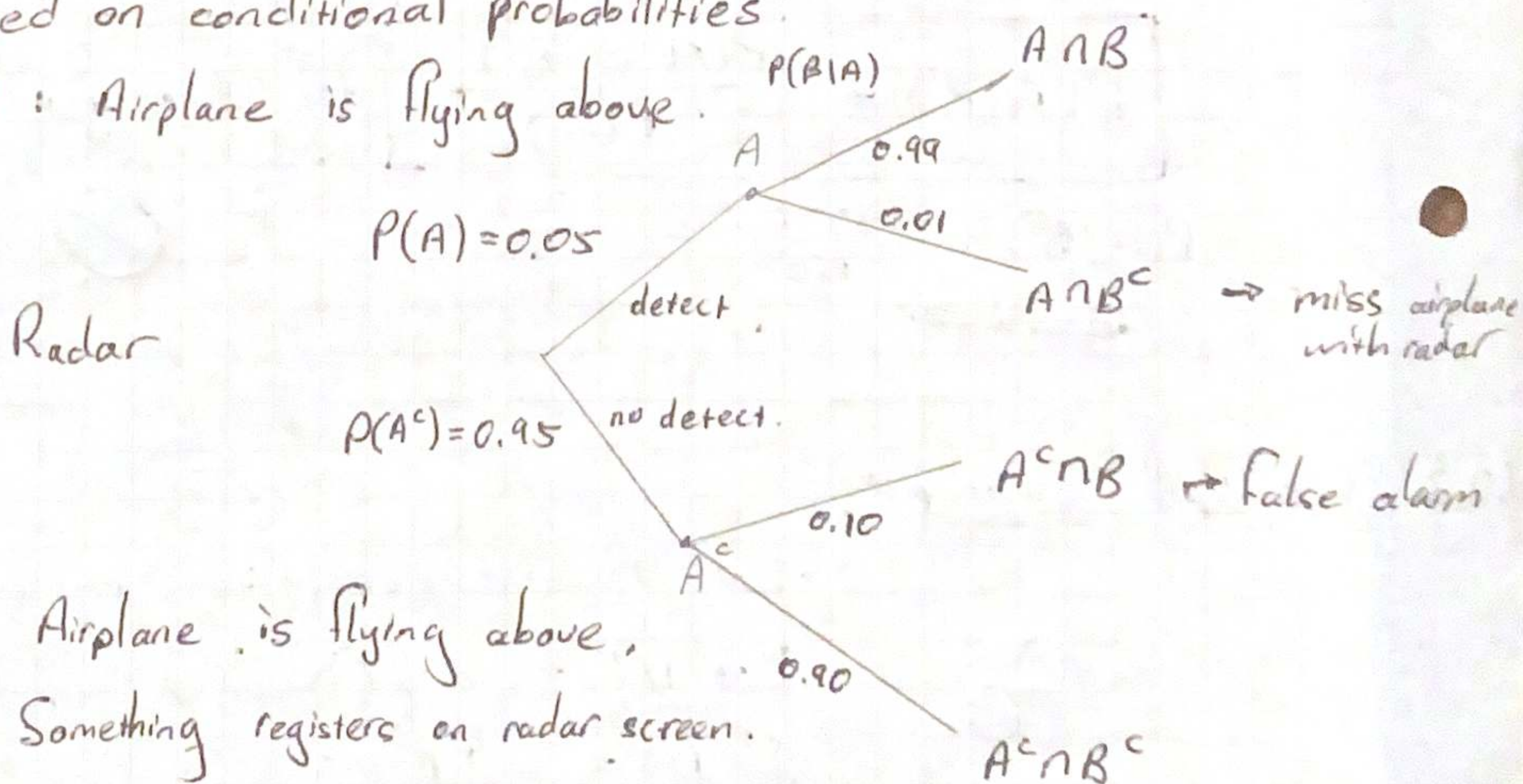
= $\frac{P(A \cap B) + P(C \cap B)}{P(B)} = \frac{P(A|B) + P(C|B)}{1}$

• also finite countable additivity.

2.5) A Radar Example and Three Basic Tools

• Models based on conditional probabilities.

• Event A : Airplane is flying above.



• Event A : Airplane is flying above.

• Event B : Something registers on radar screen.

• $P(A \cap B) = P(A) \cdot P(B|A) = 0.05 \times 0.99 = 0.0495$

• $P(B) = 0.05 \times 0.99 + 0.95 \times 0.1 = 0.1445$

• $P(A|B) = \frac{0.05 \times 0.99}{0.1445} = 0.34$

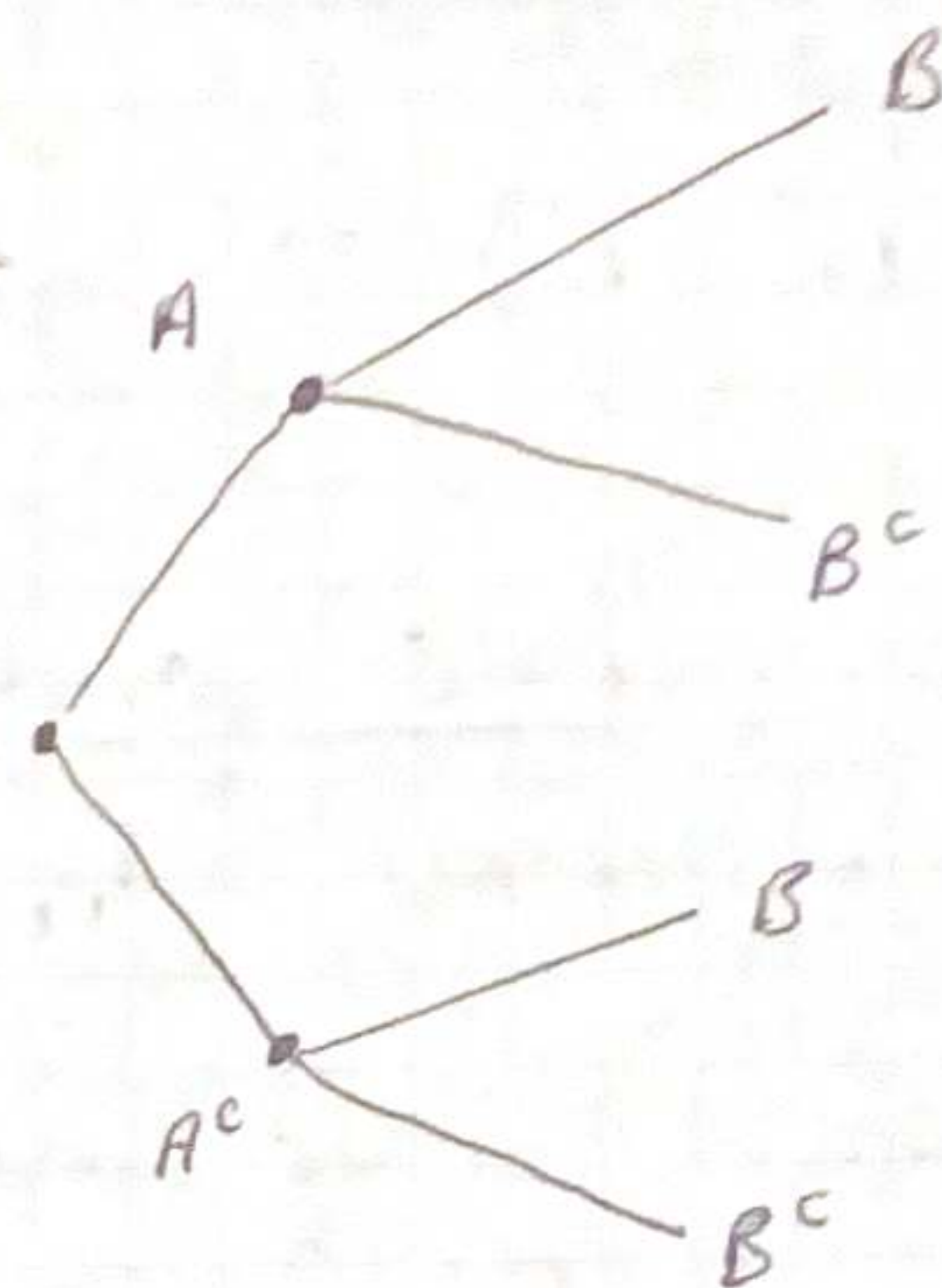
34% probability of plane actually being there given radar detects one.

L2.6 Multiplication Rule

$$P(\underline{A} | \underline{B}) = \frac{P(\underline{A} \cap \underline{B})}{P(\underline{B})}$$

$$\therefore P(\underline{A} \cap \underline{B}) = P(\underline{B}) \cdot P(\underline{A} | \underline{B})$$

$$= P(\underline{A}) \cdot P(\underline{B} | \underline{A})$$



$$\bullet P(A^c \cap B \cap C)$$

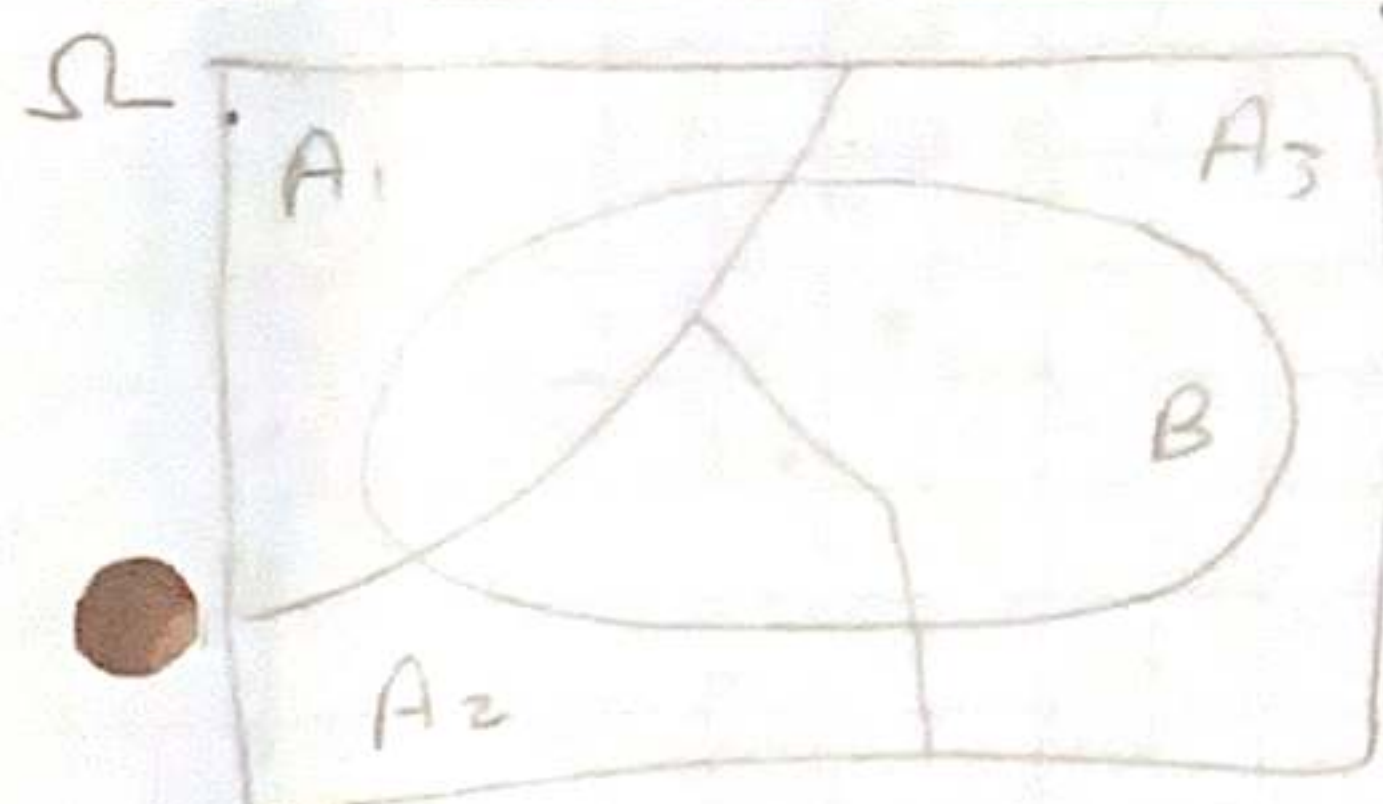
$$= P((A^c \cap B) \cap C)$$

$$= P(A^c \cap B) \cdot P(C | A^c \cap B)$$

$$= P(A^c) \cdot P(B | A^c) \cdot P(C | A^c \cap B)$$

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1) \prod_{i=2}^n P(A_i | A_1 \cap \dots \cap A_{i-1})$$

L2.7) Total Probability Theorem



• Partition of sample space into A_1, A_2, A_3
 • Have $P(A_i)$, for every i .

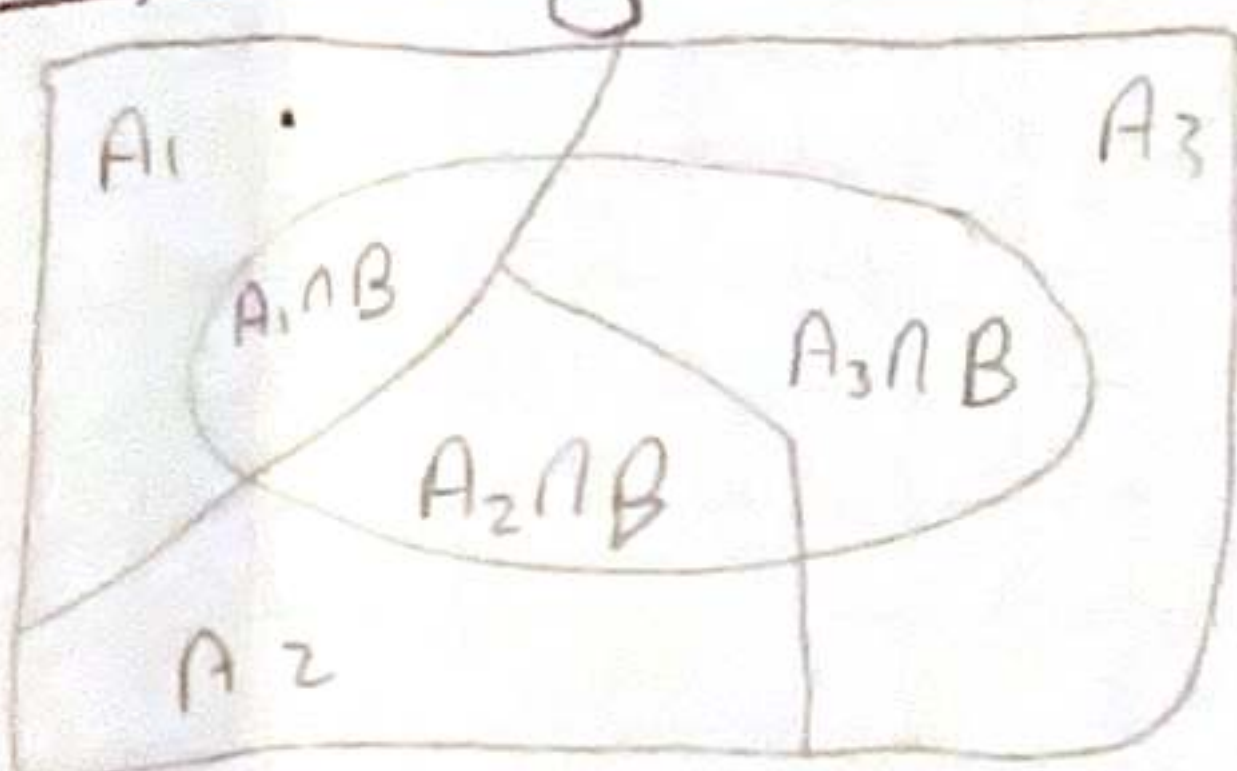
$$\bullet P(B) = P(B \cap A_1) + P(B \cap A_2) + P(B \cap A_3)$$

$$= P(A_1)P(B | A_1) + \dots + \dots$$

$$\therefore P(B) = \sum_i P(A_i)P(B | A_i)$$



L2.8) Bayes Rule



• Partition of sample space into A_1, A_2, A_3

• Have $P(A_i)$, for every i initial "beliefs"

• Have $P(B | A_i)$, for every i .

→ revised "beliefs," given that B occurred

$$P(A_i | B) = \frac{P(A_i) P(B | A_i)}{\sum_j P(A_j) P(B | A_j)}$$

- Bayes' rule and inference

→ systematic approach for incorporating new evidence.

- Bayesian Inference

- initial beliefs $P(A_i)$ on possible causes of an observed event B .
- model of the world under each A_i : $P(B | A_i)$

$$A_i \xrightarrow[P(B | A_i)]{\text{model}} B$$

- draw conclusions about causes

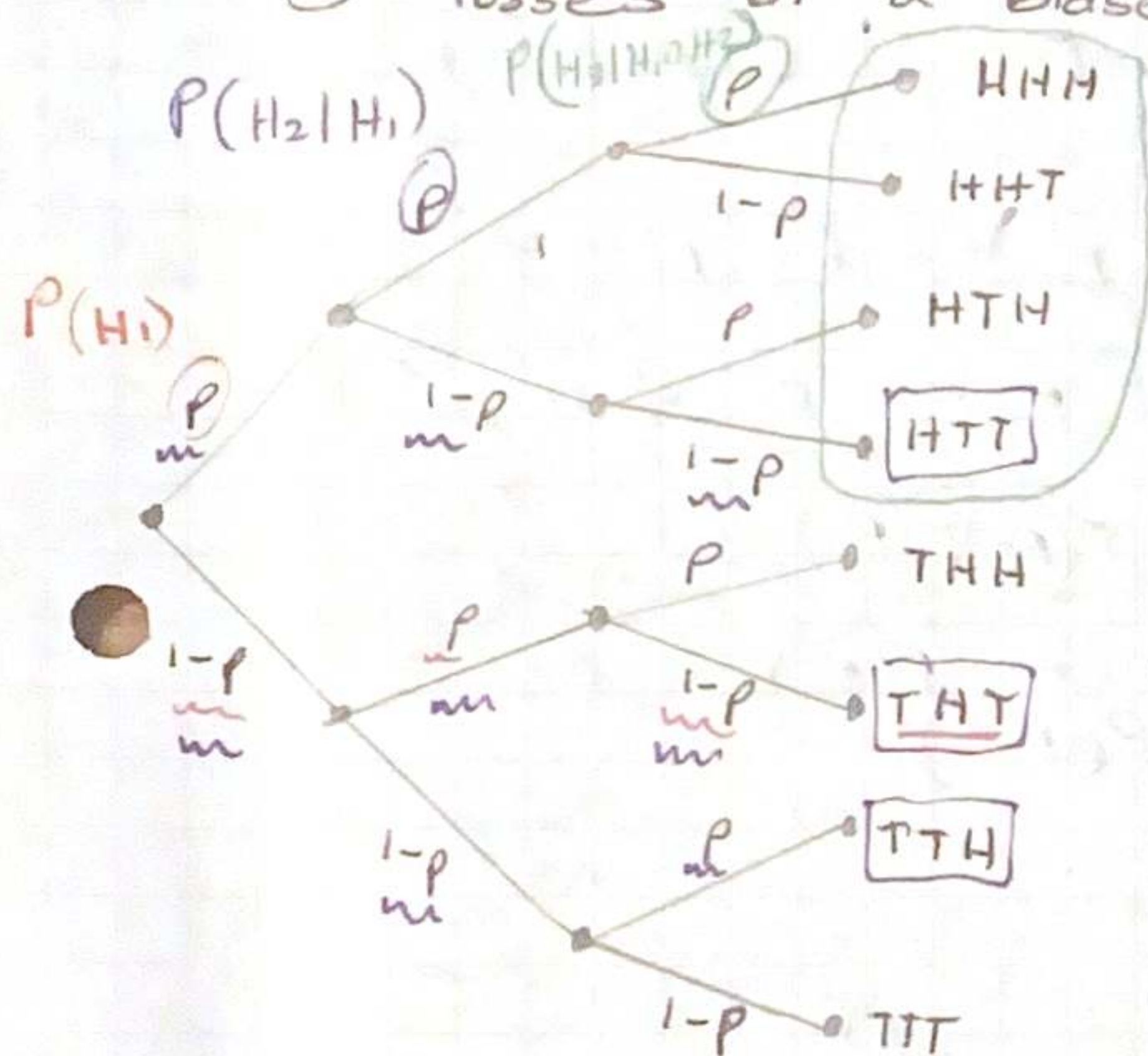
$$B \xrightarrow[P(A_i | B)]{\text{inference}} A_i$$

• Introduction to Probability - Lecture 3

L3.1) Overview of Independence

L3.2) Coin Tossing Example model based on conditional probabilities.

• 3 tosses of a biased coin: $P(H) = p$, $P(T) = 1-p$



• Multiplication rule:

~~$$P(\text{THH}) = p(1-p)p$$~~

$$P(\text{THT}) = (1-p)p(1-p)$$

• Total Probability:

$$P(\text{1 head}) = 3p(1-p)^2 \quad \left\{ \overset{1}{-} + \overset{2}{-} + \overset{3}{-} \right\}$$

• Bayes Rule:

$$P(\text{first toss is H} \mid \text{1 head})$$

$$= \frac{P(H_1 \cap \text{1 head})}{P(\text{1 head})}$$

$$= \frac{P(1-p)^2}{3p(1-p)^2} = \frac{1}{3}$$

L3.3 Independence of two Events

• Intuitive "definition": $P(B|A) = P(B)$

- occurrence of A provides no new information about B.

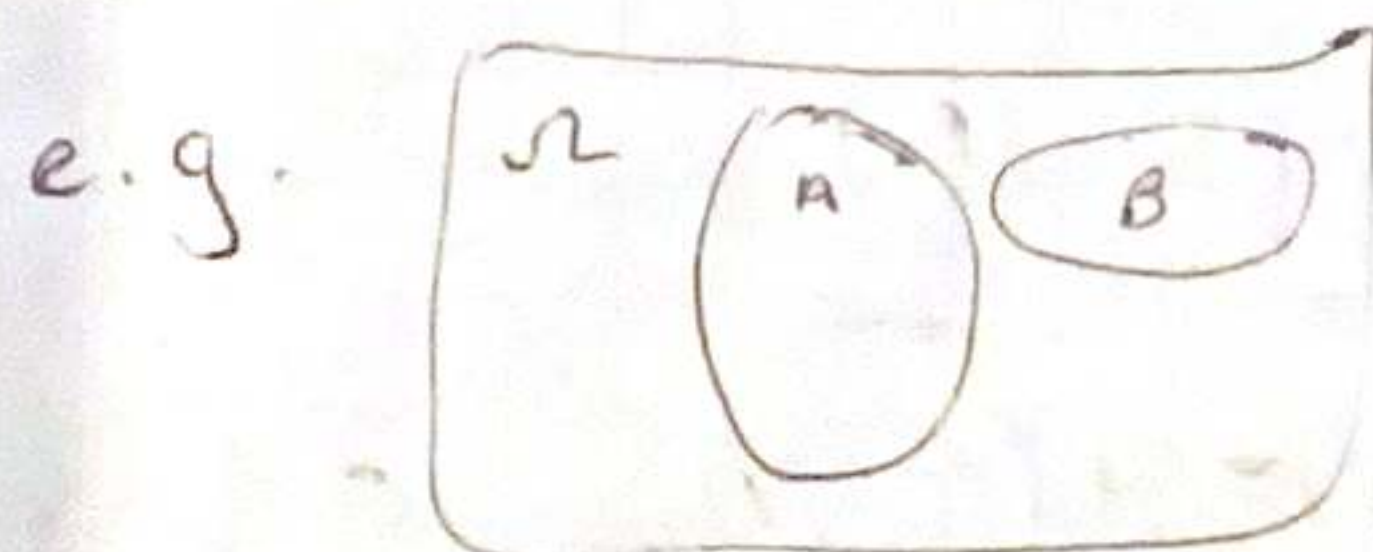
$$P(A \cap B) = P(A)P(B|A) = P(A)P(B)$$

Definition of independence: $P(A \cap B) = P(A) \cdot P(B)$

- Symmetric with respect to A and B

- implies $P(A|B) = P(A)$

- applies even if $P(A) = 0$.



$$P(A) > 0$$

$$P(B) > 0$$

Independent?

$$P(A \cap B) = 0$$

$$P(A)P(B) > 0 \therefore \text{Dependent}$$

→ Logical: If A occurs, we know B did not occur

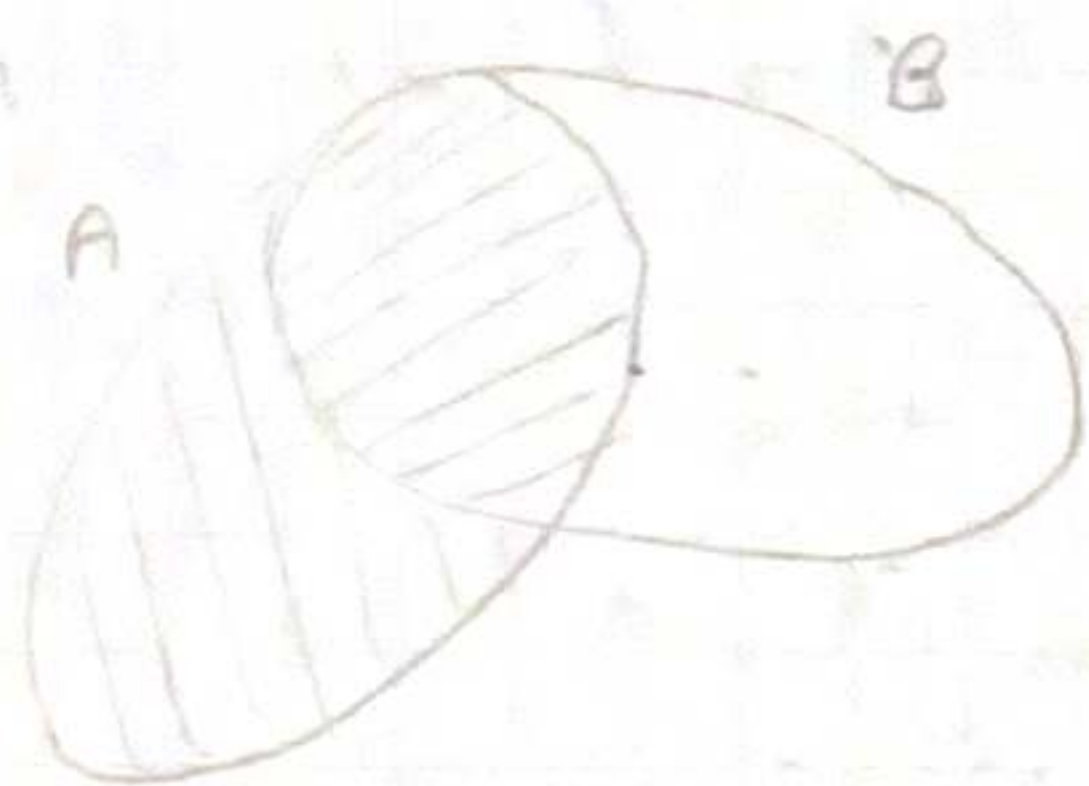
L3.4) Independence of Event Complements

Definition of Independence : $P(A \cap B) = P(A) \cdot P(B)$

If A and B are independent, then A and B^c are independent.

↳ Intuitive argument.

↳ Formal Proof



$$A = (A \cap B) \cup (A \cap B^c)$$

$$\begin{aligned} P(A) &= P(A \cap B) + P(A \cap B^c) \\ &= P(A) \cdot P(B) + P(A \cap B^c) \end{aligned}$$

$$\begin{aligned} P(A \cap B^c) &= P(A) - P(A)P(B) \\ &= P(A)(1 - P(B)) \\ &= P(A)P(B^c) \end{aligned}$$

L3.5) Conditional Independence

Conditional Independence, given C , is defined as independence under the probability law $P(\cdot | C)$.

Unconditional : $P(A \cap B) = P(A)P(B)$

Conditional : $P(A \cap B | C) = P(A | C) \cdot P(B | C)$

Does one type of independence imply the other?

↳ Independence does not imply conditional independence

↳ NO.

L3.6) Independence vs. Conditional Independence

↳ They are not the same. Separate types of independence.

↳ Check ~~u~~ Can be related +

$$\left. \begin{aligned} P(H | \text{coin } A) &= 0.9 \\ P(H | \text{coin } B) &= 0.1 \end{aligned} \right\} \text{Unfair Coin Flip}$$

Compare:

$$\begin{aligned} P(\text{toss 11} = H) &= P(A)P(H_{11} | A) + P(B)P(H_{11} | B) \\ &= 0.5 \times 0.9 + 0.5 \times 0.1 = \underline{0.5} \end{aligned}$$

$$P(\text{toss 11} = H | \text{first 10 tosses are heads}) = P(H_{11} | A) = 0.9$$

↳ Only possible if coin $p \rightarrow$ change opinion of expected outcome

L3.7) Independence of a Collection of Events

- Independence $\rightarrow$ many events $\rightarrow$ all independent?
- Intuition $\rightarrow$ Info on some events doesn't change $P()$ of remaining events

Definition: Events $A_1, A_2, \dots, A_n$ are called independent if:
 $P(A_i \cap A_j \cap \dots \cap A_m) = P(A_i)P(A_j) \dots P(A_m)$ for any distinct indices $i, j, \dots, m$.

$n = 3$:

$$\left. \begin{aligned} P(A_1 \cap A_2) &= P(A_1)P(A_2) \\ P(A_1 \cap A_3) &= P(A_1)P(A_3) \\ P(A_2 \cap A_3) &= P(A_2)P(A_3) \end{aligned} \right\} \text{pairwise independence.}$$

$$P(A_1 \cap A_2 \cap A_3) = P(A_1) \cdot P(A_2) \cdot P(A_3)$$

L3.8) Independence Versus Pairwise Independence

HH	HT
TH	TT

C

$$P(H_1 \cap C) = P(H_1 \cap H_2) = 1/4$$

$$P(H_1)P(C) = 1/2 \cdot (1/4 + 1/4) = 1/4$$

H_1, C : independent
 H_2, C : independent.

$$\left. \begin{aligned} P(H_1 \cap H_2 \cap C) &= P(HH) = 1/4 \\ P(H_1)P(H_2)P(C) &= 1/8 \end{aligned} \right\} \text{Difference !!}$$

$\hookrightarrow$ Independent H_1, H_2, C are pairwise independent,
 but not independent.

In contrast:

$$P(C | H_1) = P(H_2 | H_1) = P(H_2) = 1/2 = P(C)$$

$\hookrightarrow$ event having occurred does not inform you of what to expect for the next event.

$$P(C | H_1 \cap H_2) = 1 \neq P(C) = 1/2$$

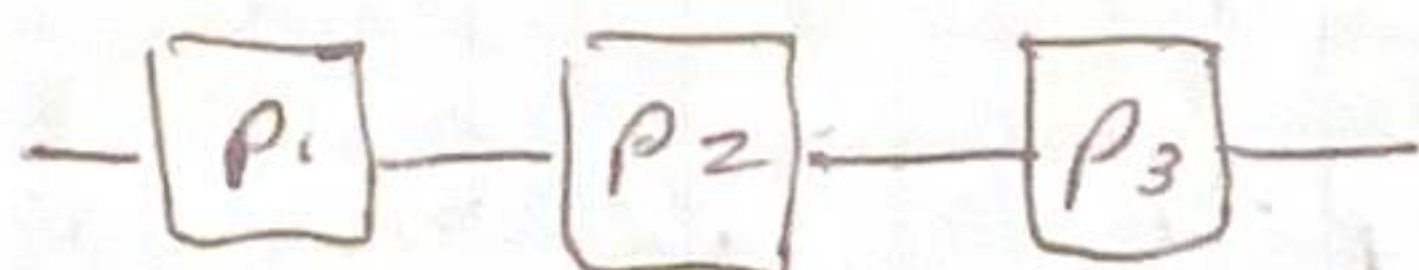
$\hookrightarrow$ Together H_1 & H_2 are not independent to C
 $\hookrightarrow$ So as a collection of events H_1, H_2 & C are not indep.

L3.9) Reliability

p_i : probability that unit i is "up"
independent units.

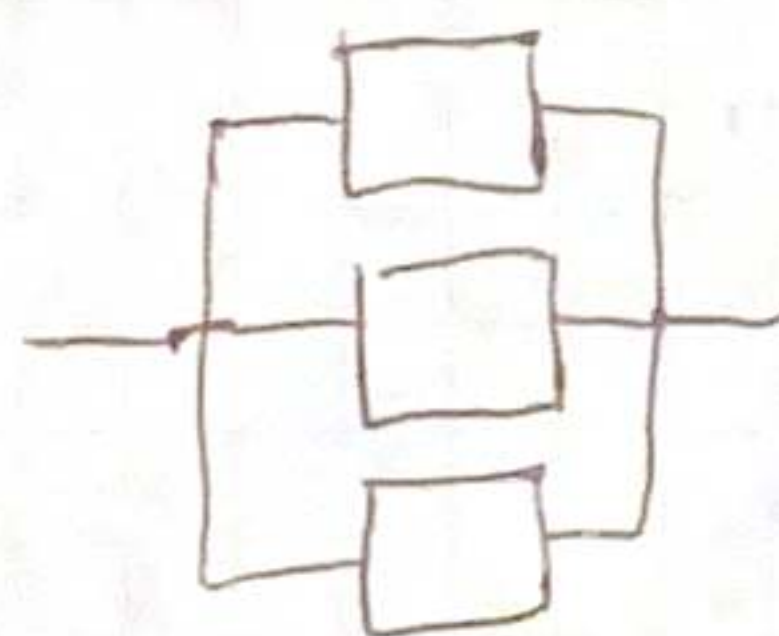
U_i : i th unit ~~is~~ up
 $U_1, U_2, \dots, U_m$ independent

F_i : i th unit down
 $\Rightarrow F_i$ independent.



$$\begin{aligned}
 P(\text{system up}) &= P(U_1 \cap U_2 \cap U_3) \\
 &= P(U_1) \cdot P(U_2) \cdot P(U_3) \\
 &= p_1 \cdot p_2 \cdot p_3
 \end{aligned}$$

~~$P(\text{system is up}) = P(U_1 \cup U_2 \cup U_3)$~~ ← De Morgan's law used here!



$$\begin{aligned}
 P(\text{system is up}) &= P(U_1 \cup U_2 \cup U_3) \\
 &= 1 - P(F_1 \cap F_2 \cap F_3) \\
 &= 1 - P(F_1)P(F_2)P(F_3) \\
 &= 1 - (1-p_1)(1-p_2)(1-p_3)
 \end{aligned}$$

L3.10) The King's Sibling

King comes from a family of two children.

What is the probability that his sibling is female? ①

Assume: boy, have precedence

$$P(\text{boy}) = P(\text{girl}) = 1/2 \quad \text{independent}$$

① $P(\text{girl}) = 1/2$? correct?

$1/4$ BB	$1/4$ BG
$1/4$ GB	$1/4$ GG

→ Know there is a "king"

$\therefore$ Equal chance $\rightarrow P(\text{boy}) = P(\text{girl}) = 1/3$

$\therefore P(\text{girl}) = 1/3 + 1/3 = 2/3$? correct?

What if have children till 1 boy?
P(1 boy) = 1/2

Problem Set Observations - Section 1.3 Textbook.

• Sequential Conditioning using the Multiplication Rule

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1) \cdot P(A_2 | A_1) \cdot P(A_3 | A_1 \cap A_2) \dots$$

↳ Items drawn one at a time without replacement, and you care about a sequence of outcomes all going a certain way.

Strategy:

1. Define success as an intersection
↳ all draws must be non-defective:

$$A = A_1 \cap A_2 \cap A_3 \cap A_4$$

2. At each step, update the population
↳ both the pool size and defect outcome count shrink as items are drawn.

3. Multiply the conditional probabilities sequentially.

<u>Draw</u>	<u>Non-defective Remaining</u>	<u>Pool / Population Remaining</u>
1st	95	100
2nd	94	99
3rd	93	98
4th	92	97

~~Alternatively~~

Alternatively:

$$\frac{\binom{\text{non-defective population}}{\text{defective draws}}}{\binom{\text{total population}}{\text{draws}}} = \frac{\binom{95}{4}}{\binom{100}{4}}$$

↳ Numerator → ways to choose #draws items from non-defective ones

↳ Denominator → total ways to choose #draws from total population